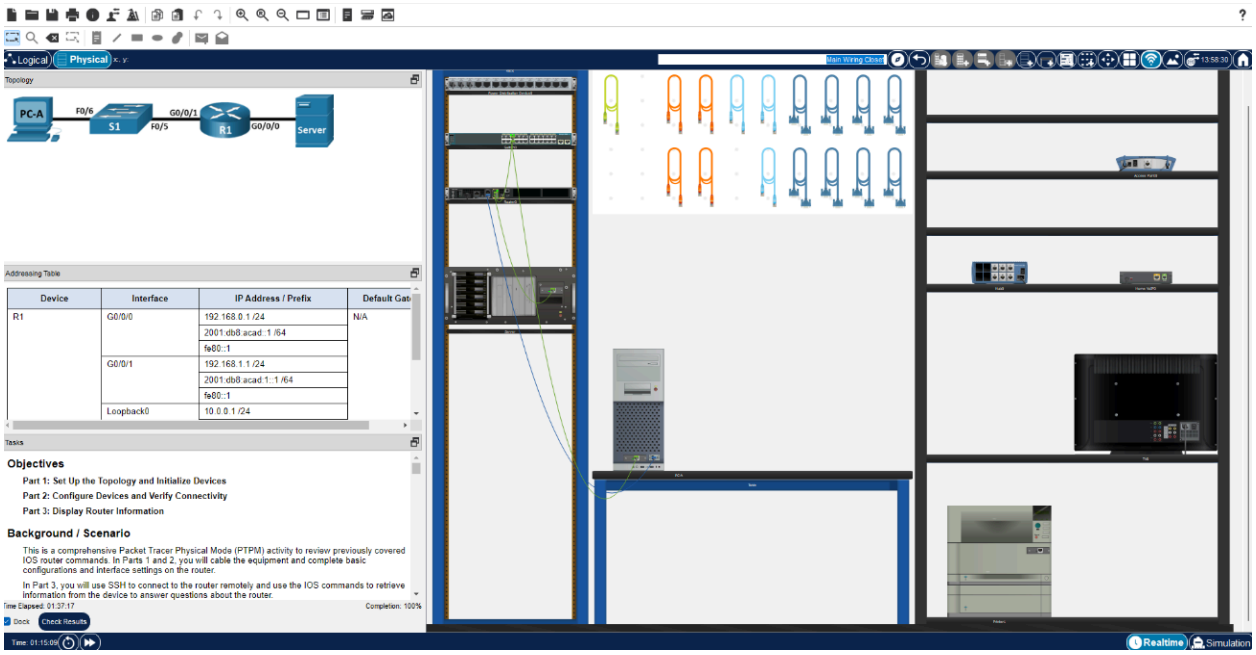
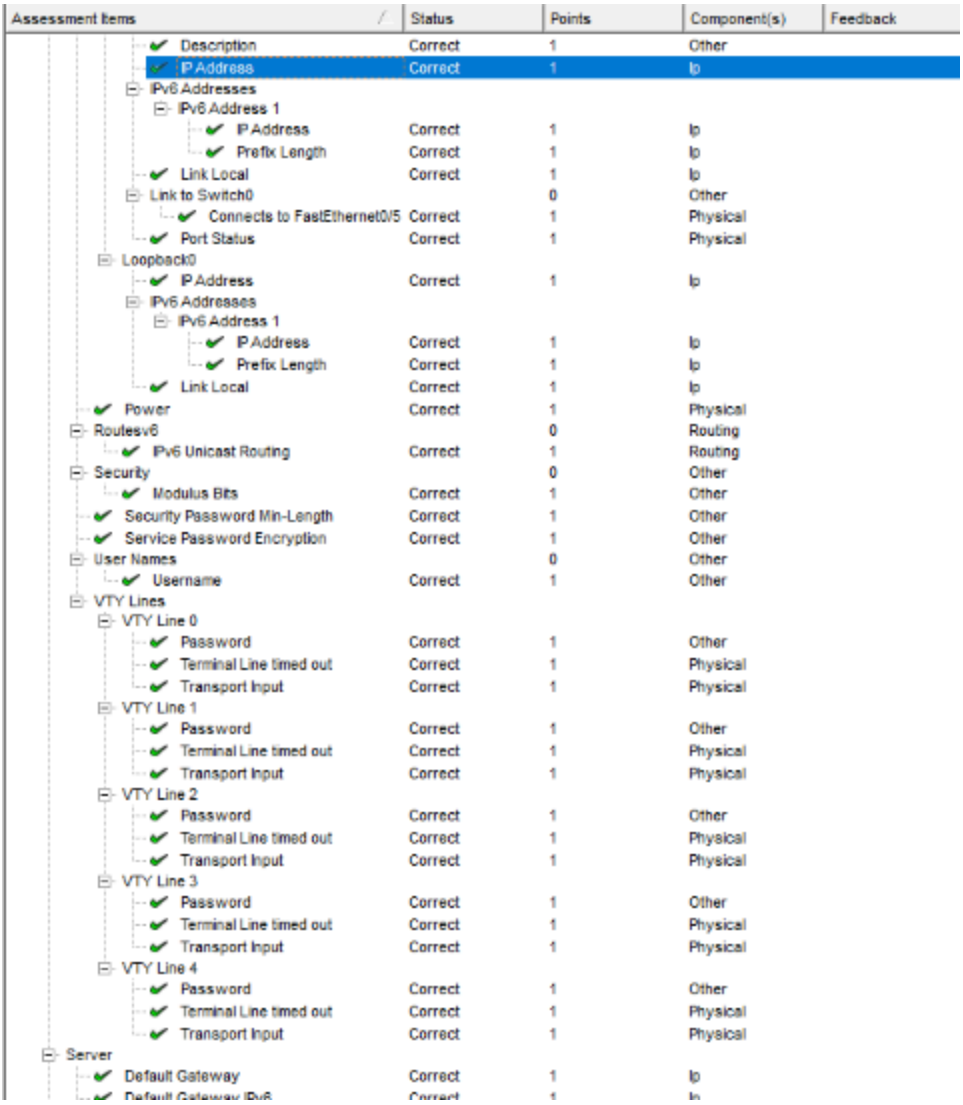
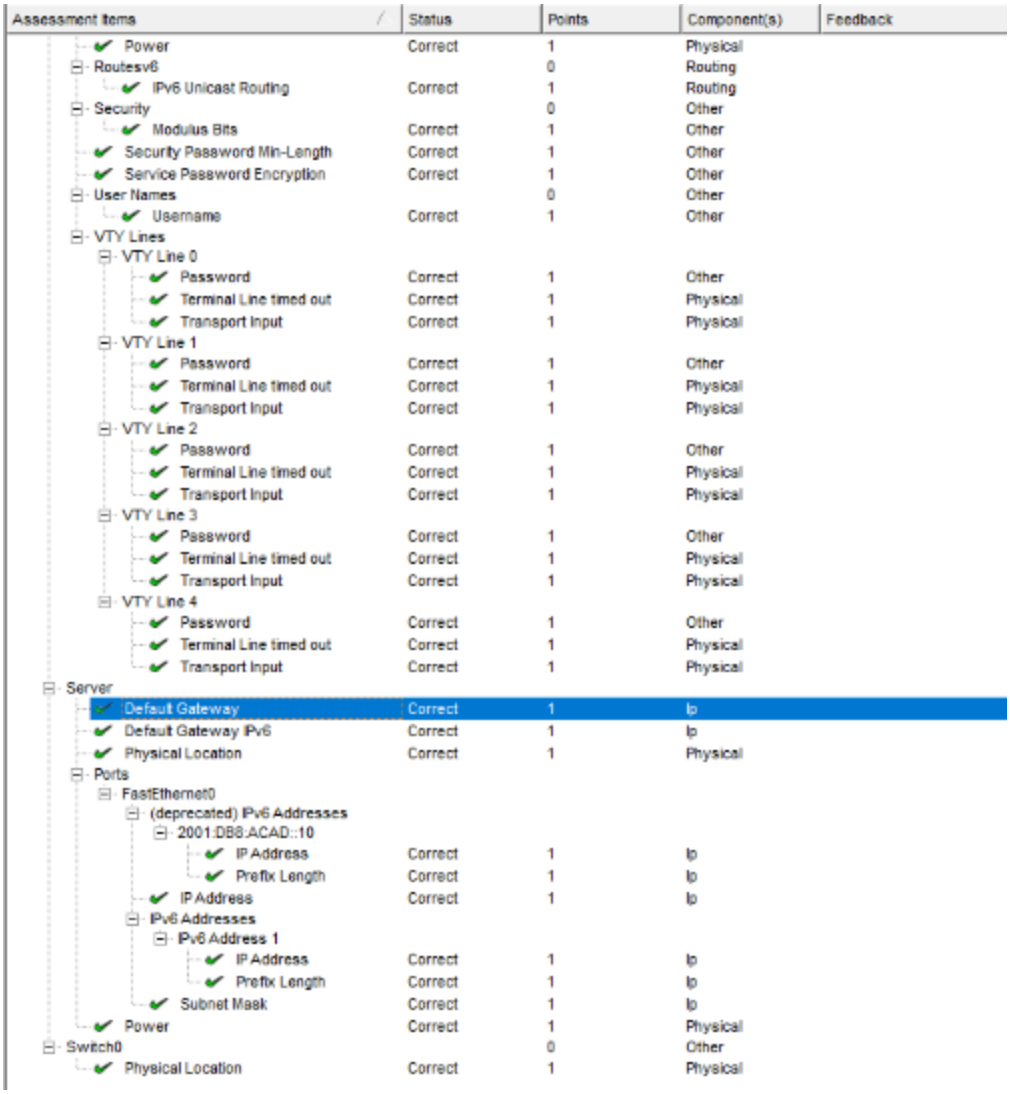

Logical
Physical
Topology
PC-A
F0/6
S1
F0/5
G0/0/1
R1
G0/0/0
Server
Addressing Table
Device
Interface
IP Address / Prefix
Default Gat
R1
G0/0/0
192.168.0.1 /24
N/A
2001:db8:acad::1 /64
fe80::1
G0/0/1
192.168.1.1 /24
2001:db8:acad:1::1 /64
fe80::1
Loopback0
10.0.0.1 /24
Tasks
Objectives
Part 1: Set Up the Topology and Initialize Devices
Part 2: Configure Devices and Verify Connectivity
Part 3: Display Router Information
Background / Scenario
This is a comprehensive Packet Tracer Physical Mode (PTPM) activity to review previously covered IOS router commands. In Parts 1 and 2, you will cable the equipment and complete basic configurations and interface settings on the router.
In Part 3, you will use SSH to connect to the router remotely and use the IOS commands to retrieve information from the device to answer questions about the router.
Time Elapsed: 01:37:17
Completion: 100%
Dock
Check Results
Time: 01:15:09
Realtime
Simulation

Assessment Items	Status	Points	Component(s)	Feedback
Network				
PC-A				
Default Gateway	Correct	1	Ip	
Default Gateway IPv6	Correct	1	Ip	
Physical Location	Correct	1	Physical	
Ports				
FastEthernet0				
IP Address	Correct	1	Ip	
IPv6 Addresses				
IPv6 Address 1				
IP Address	Correct	1	Ip	
Prefix Length	Correct	1	Ip	
Link to Switch0		0	Other	
Connects to FastEthernet0/6	Correct	1	Physical	
Subnet Mask	Correct	1	Ip	
Power	Correct	1	Physical	
RS 232		0	Other	
Link to Router0		0	Other	
Connects to Console	Correct	1	Physical	
Router0				
Banner MOTD	Correct	1	Other	
Console Line				
Login	Correct	1	Physical	
Password	Correct	1	Other	
Terminal Line timed out	Correct	1	Physical	
Enable Secret	Correct	1	Other	
Host Name	Correct	1	Other	
IP Domain Name	Correct	1	Other	
Login Options				
Blocking				
Attempts	Correct	1	Other	
Duration	Correct	1	Other	
Enabled	Correct	1	Other	
Period	Correct	1	Other	
Physical Location	Correct	1	Physical	
Ports				
GigabitEthernet0/0/0				
Description	Correct	1	Other	
IP Address	Correct	1	Ip	
IPv6 Addresses				
IPv6 Address 1				
IP Address	Correct	1	Ip	
Prefix Length	Correct	1	Ip	
Link Local	Correct	1	Ip	
Link to Server		0	Other	
Connects to FastEthernet0	Correct	1	Physical	
Port Status	Correct	1	Physical	
GigabitEthernet0/0/1				
Description	Correct	1	Other	
IP Address	Correct	1	Ip	



Assessment Items	Status	Points	Component(s)	Feedback
Description	Correct	1	Other	
IP Address	Correct	1	Ip	
IPv6 Addresses				
IPv6 Address 1				
IP Address	Correct	1	Ip	
Prefix Length	Correct	1	Ip	
Link Local	Correct	1	Ip	
Link to Switch0		0	Other	
Connects to FastEthernet0/5	Correct	1	Physical	
Port Status	Correct	1	Physical	
Loopback0				
IP Address	Correct	1	Ip	
IPv6 Addresses				
IPv6 Address 1				
IP Address	Correct	1	Ip	
Prefix Length	Correct	1	Ip	
Link Local	Correct	1	Ip	
Power	Correct	1	Physical	
Routesv6		0	Routing	
IPv6 Unicast Routing	Correct	1	Routing	
Security		0	Other	
Modulus Bits	Correct	1	Other	
Security Password Min-Length	Correct	1	Other	
Service Password Encryption	Correct	1	Other	
User Names		0	Other	
Username	Correct	1	Other	
VTY Lines				
VTY Line 0				
Password	Correct	1	Other	
Terminal Line timed out	Correct	1	Physical	
Transport Input	Correct	1	Physical	
VTY Line 1				
Password	Correct	1	Other	
Terminal Line timed out	Correct	1	Physical	
Transport Input	Correct	1	Physical	
VTY Line 2				
Password	Correct	1	Other	
Terminal Line timed out	Correct	1	Physical	
Transport Input	Correct	1	Physical	
VTY Line 3				
Password	Correct	1	Other	
Terminal Line timed out	Correct	1	Physical	
Transport Input	Correct	1	Physical	
VTY Line 4				
Password	Correct	1	Other	
Terminal Line timed out	Correct	1	Physical	
Transport Input	Correct	1	Physical	
Server				
Default Gateway	Correct	1	Ip	
Default Gateway IPv6	Correct	1	Ip	



Assessment Items	Status	Points	Component(s)	Feedback
Power	Correct	1	Physical	
Routesv6		0	Routing	
IPv6 Unicast Routing	Correct	1	Routing	
Security		0	Other	
Modulus Bits	Correct	1	Other	
Security Password Min-Length	Correct	1	Other	
Service Password Encryption	Correct	1	Other	
User Names		0	Other	
Username	Correct	1	Other	
VTY Lines				
VTY Line 0				
Password	Correct	1	Other	
Terminal Line timed out	Correct	1	Physical	
Transport Input	Correct	1	Physical	
VTY Line 1				
Password	Correct	1	Other	
Terminal Line timed out	Correct	1	Physical	
Transport Input	Correct	1	Physical	
VTY Line 2				
Password	Correct	1	Other	
Terminal Line timed out	Correct	1	Physical	
Transport Input	Correct	1	Physical	
VTY Line 3				
Password	Correct	1	Other	
Terminal Line timed out	Correct	1	Physical	
Transport Input	Correct	1	Physical	
VTY Line 4				
Password	Correct	1	Other	
Terminal Line timed out	Correct	1	Physical	
Transport Input	Correct	1	Physical	
Server				
Default Gateway	Correct	1	Ip	
Default Gateway IPv6	Correct	1	Ip	
Physical Location	Correct	1	Physical	
Ports				
FastEthernet0				
(deprecated) IPv6 Addresses				
2001:DB8:ACAD::10				
IP Address	Correct	1	Ip	
Prefix Length	Correct	1	Ip	
IP Address	Correct	1	Ip	
IPv6 Addresses				
IPv6 Address 1				
IP Address	Correct	1	Ip	
Prefix Length	Correct	1	Ip	
Subnet Mask	Correct	1	Ip	
Power	Correct	1	Physical	
Switch0		0	Other	
Physical Location	Correct	1	Physical	

คำถามที่ 1: ในการตรวจสอบปัญหาการเชื่อมต่อเครือข่าย ช่างเทคนิคสงสัยว่าอินเทอร์เฟซไม่ได้ถูกเปิดใช้งาน (enabled) ช่างเทคนิคควรใช้คำสั่ง show ใดในการตรวจสอบปัญหานี้?
คำตอบ: ช่างเทคนิคสามารถใช้คำสั่ง show ip interface brief หรือ show interfaces เพื่อตรวจสอบสถานะของอินเทอร์เฟซ หากอินเทอร์เฟซไม่ได้ถูกเปิดใช้งาน สถานะของ Interface จะแสดงเป็น "administratively down"
คำถามที่ 2: ในการตรวจสอบปัญหาการเชื่อมต่อเครือข่าย ช่างเทคนิคสงสัยว่าอินเทอร์เฟซถูกกำหนดค่า Subnet Mask ไม่ถูกต้อง ช่างเทคนิคควรใช้คำสั่ง show ใดในการตรวจสอบปัญหานี้?
คำตอบ: ช่างเทคนิคสามารถใช้คำสั่ง show ip interface เพื่อตรวจสอบรายละเอียดการกำหนดค่า IP Address และ Subnet Mask ของอินเทอร์เฟซนั้นๆ เพื่อยืนยันว่าถูกต้องหรือไม่